

APPENDIX V

Some Important Plasma Parameters

1. Electron plasma frequency:

$$\omega_{pe} = \left(\frac{n_e e^2}{m_e \epsilon_0} \right)^{1/2} = 56.5 n_e^{1/2} \text{ rad/sec } (n_e \text{ in } m^{-3})$$

2. Ion plasma frequency:

$$\omega_{pi} = \left(\frac{n_i Z^2 e^2}{m_i \epsilon_0} \right)^{1/2}$$

3. Debye length:

$$\lambda_D = \left(\frac{\epsilon_0 k T}{n_e e^2} \right)^{1/2} = 69.0 \left(\frac{T}{n_e} \right)^{1/2} \text{ m}$$

(T in degrees K and n_e in m^{-3})

4. Electron cyclotron frequency:

$$\omega_{ce} = \frac{eB}{m_e} = 1.76 \times 10^{11} B \text{ rad/sec } (B \text{ in tesla})$$

5. Ion cyclotron frequency:

$$\omega_{ci} = \frac{Z e B}{m_i}$$

6. Particle magnetic moment:

$$\vec{m} = - \frac{W_{\perp}}{B^2} \vec{B} = - \frac{m v_{\perp}^2 / 2}{B^2} \vec{B}$$

7. Electron cyclotron radius:

$$r_{ce} = \frac{v_{\perp e}}{\omega_{ce}} = \frac{m_e v_{\perp e}}{e B}$$

8. Ion cyclotron radius:

$$r_{ci} = \frac{v_{\perp i}}{\omega_{ci}} = \frac{m_i v_{\perp i}}{Z e B}$$

9. Number of electrons in Debye sphere:

$$N_D = \frac{4}{3} \pi \lambda_D^3 n_e = 1.37 \times 10^6 \frac{T^{3/2}}{n_e^{1/2}}$$

(T in degrees K and n_e in m^{-3})

10. Alfvén velocity:

$$\vec{v}_A = \frac{\vec{B}}{(\mu_0 \rho_m)^{1/2}}$$

11. DC conductivity:

$$\sigma_o = \frac{n_e e^2}{m_e v_e}$$

12. Free electron diffusion coefficient:

$$D_e = \frac{k T_e}{m_e v_e}$$

13. Ambipolar diffusion coefficient:

$$D_a = \frac{k(T_e + T_i)}{m_e v_{en} + m_i v_{in}}$$

14. Magnetic pressure:

$$p_m = \frac{B^2}{2\mu_0}$$

15. Magnetic viscosity:

$$\eta_m = \frac{1}{\mu_0 \sigma_0}$$

16. Magnetic Reynolds number:

$$R_m = \frac{uL}{\eta_m}$$

17. Coulomb cutoff parameter:

$$\begin{aligned} \Lambda &= 12 \pi n_e \lambda_D^3 = 9 N_D \\ &= 1.23 \times 10^7 \frac{T^{3/2}}{n_e^{1/2}} \end{aligned}$$

(T in degrees K and n_e in m^{-3})

18. Electron collision frequencies for momentum transfer:

$$\nu_{ei} = 3.62 \times 10^{-6} n_i T_e^{-3/2} \ln \Lambda \text{ sec}^{-1}$$

$$\nu_{en} = 2.60 \times 10^4 \sigma^2 n_n T_e^{1/2} \text{ sec}^{-1}$$

(T in degrees K, $n_{i,n}$ in m^{-3} ; σ is the sum of the radii of the colliding particles and is of the order of 10^{-10} m, and $\ln \Lambda$ is typically about 10).